

Carter Jensen

GIS and Computer Science Student

Portfolio: carterjen3ct.github.io

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EDUCATION

James Madison University — Harrisonburg, Virginia

Expected Graduation: Spring 2027

- B.S., Geography, Concentration: GIS, GPA: 3.5 | Minor: Computer Science, GPA: 3.7 | Overall GPA: 3.5
- Dean's List Honors: 2023, 2024, 2025
- Gamma Theta Upsilon Geography Honor Society
- James Madison A.I. Club
- Geographic Angles on Public Opinion Research Lab
- JMU 2027 Hydropower Collegiate Competition Team

CERTIFICATIONS

- Esri Technical Certification — ArcGIS Pro Associate, Certified September 2026

PROJECTS

Flood Inundation Mapping Web App — *Google Earth Engine, JavaScript, Sentinel-1 SAR, Sentinel-2*

- Built an interactive GEE app mapping flood extent from Sentinel-1 SAR backscatter and Sentinel-2 NDWI, validated against Pakistan's Sindh Province 2022 flood event
- Performed accuracy assessment using stratified random sampling (100 points) and confusion matrix analysis to quantify classification reliability

Sidewalk Suitability Analysis — *ArcGIS Pro, Geoprocessing, Spatial Analysis*

- Conducted weighted-overlay suitability analysis to identify optimal new sidewalk locations in Harrisonburg, VA
- Integrated multiple datasets with point density analysis, standardized raster layers, buffer, and masking geoprocessing tools via the ArcGIS raster calculator

EXPERIENCE

Team Lead / Site-Operations Coordinator — Abiding Presence Lutheran Church, Burke, VA 2021 – 2026

- Solely responsible for planning and leading weeklong camps at 5 different locations across the DC Metro Area
- Coordinated teams of 12+ staff at each location and led daily programs for 100+ children
- Programs success led to an expanded grant from the Metro DC Synod to grow the program

SKILLS / SOFTWARE

GIS: ArcGIS Pro, ArcGIS Online, ArcMap, ArcGIS StoryMaps, QGIS

Remote Sensing: Google Earth Engine, ENVI

Programming: Python (NumPy, Pandas, ArcPy, Pillow), Java; HTML, CSS, JavaScript; AI fluency

Other: Adobe Illustrator

RELEVANT COURSEWORK

Programming for GIS • **Intermediate Geographic Information Science** • Applications of GIS (Urban Planning) • Digital Image Processing & Remote Sensing • **Cloud-Based Remote Sensing (GEE)** • Earth Observation & Satellite Navigation Systems • Cartography & Geospatial Visualization • Application Development • **Software Engineering** • Urban Analytics in R • Statistics & Calculus • **Advanced Cartography** • Discrete Structures • **Web Development** • **Applications of GIS (Advanced Geospatial Tools)**